

4-Bit Signed ALU - Arduino Display System

This document describes the Arduino display system used with the completed ALU. The Arduino is a display/readout helper only. It does not calculate the ALU results.

Purpose

The physical ALU produces many output bits: arithmetic sum, product, comparator flags, overflow, and control states. Reading all of these only with LEDs can be hard, so the Arduino collects the output bits through shift registers and prints the selected result on an LCD.

Hardware Interface

ALU outputs -> 74HC165 parallel inputs
74HC165 serial output -> Arduino Nano
Arduino Nano -> LCD1602 I2C display

Signal Mapping

Signal group	Signals
ADD/SUB	S0, S1, S2, S3, Cout, Overflow
Comparator	GT, EQ, LT
Multiplier	FinalProduct P0-P7
Controls	SUB, SEL0, SEL1

Why 74HC165 Shift Registers Are Used

The Arduino Nano does not have enough convenient pins to read every ALU output directly. The 74HC165 shift registers allow many parallel signals to be loaded at once and then shifted into the Arduino using only a few pins.

LOAD pulse: capture all ALU output bits
CLK pulses: shift bits one by one
DATA pin: Arduino reads the serial bit stream

Display Behavior

The LCD shows the operation selected by SEL1 and SEL0. For ADD/SUB, it displays the 4-bit result and overflow flag. For comparison, it displays GT, EQ, or LT. For multiplication, it displays the 8-bit signed product.

Important Design Note

The Arduino code only formats and displays values already produced by the hardware. This keeps the project honest as a logic IC ALU. The Arduino is treated like a display module, not as a processor doing the arithmetic.